

[MS-GPNAP]:

Group Policy: Network Access Protection (NAP) Extension

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1 Introduction

The Group Policy: Network Access Protection (NAP) Extension protocol specifies functionality to control client computer access to network resources. Access can be granted or restricted per client computer based on its identity and its degree of compliance with corporate governance policy. For non-compliant client computers, NAP specifies automatic methods to reinstate compliance and to dynamically upgrade access to network resources.

Sections 1.7 and 2 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

Active Directory domain: A **domain** hosted on Active Directory. For more information, see [\[MS-ADTS\]](#).

certification authority (CA): A third party that issues **public key** certificates. Certificates serve to bind public keys to a user identity. Each user and certification authority (CA) can decide whether to trust another user or CA for a specific purpose, and whether this trust is to be transitive. For more information, see [\[RFC3280\]](#).

client-side extension GUID (CSE GUID): A **GUID** that enables a specific client-side extension on the Group Policy client to be associated with policy data that is stored in the logical and physical components of a **Group Policy Object (GPO)** on the Group Policy server, for that particular extension.

cryptographic service provider (CSP): A software module that implements cryptographic functions for calling applications that generates digital signatures. Multiple CSPs can be installed. A CSP is identified by a name represented by a NULL-terminated Unicode string.

domain: A set of users and computers sharing a common namespace and management infrastructure. At least one computer member of the set has to act as a **domain controller (DC)** and host a member list that identifies all members of the domain, as well as optionally hosting the Active Directory service. The domain controller provides authentication of members, creating a unit of trust for its members. Each domain has an identifier that is shared among its members. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5 and [\[MS-ADTS\]](#).

domain controller (DC): The service, running on a server, that implements Active Directory, or the server hosting this service. The service hosts the data store for objects and interoperates with other **DCs** to ensure that a local change to an object replicates correctly across all **DCs**. When Active Directory is operating as Active Directory Domain Services (AD DS), the **DC** contains full NC replicas of the configuration naming context (config NC), schema naming context (schema NC), and one of the domain NCs in its forest. If the AD DS **DC** is a global catalog server (GC server), it contains partial NC replicas of the remaining domain NCs in its forest. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5.2 and [\[MS-ADTS\]](#). When Active Directory is operating as Active Directory Lightweight Directory Services (AD LDS), several AD LDS **DCs** can run on one server. When Active Directory is operating as AD DS, only one AD DS **DC** can run on one server. However, several AD LDS **DCs** can coexist with one AD DS **DC** on one server. The AD LDS **DC** contains full NC replicas of the config NC and the schema NC in its forest. The domain controller is the server side of Authentication Protocol Domain Support [\[MS-APDS\]](#).

Dynamic Host Configuration Protocol (DHCP): A protocol that provides a framework for passing configuration information to hosts on a TCP/IP network, as described in [\[RFC2131\]](#).

enforcement client: An **enforcement client** uses the health state of a computer to request a certain level of access to a network. For more information about **enforcement clients**, see [\[MSDN-NAP\]](#).

globally unique identifier (GUID): A term used interchangeably with universally unique identifier (UUID) in Microsoft protocol technical documents (TDs). Interchanging the usage of these terms does not imply or require a specific algorithm or mechanism to generate the value. Specifically, the use of this term does not imply or require that the algorithms described in [\[RFC4122\]](#) or [\[C706\]](#) have to be used for generating the GUID. See also universally unique identifier (UUID).

Group Policy: A mechanism that allows the implementer to specify managed configurations for users and computers in an Active Directory service environment.

Group Policy Object (GPO): A collection of administrator-defined specifications of the policy settings that can be applied to groups of computers in a domain. Each GPO includes two elements: an object that resides in the Active Directory for the domain, and a corresponding file system subdirectory that resides on the sysvol DFS share of the Group Policy server for the domain.

Group Policy server: A server holding a database of **Group Policy Objects (GPOs)** that can be retrieved by other machines. The Group Policy server has to be a **domain controller (DC)**.

health certificate enrollment agent (HCEA): The client-side component in the Health Certificate Enrollment Protocol. The **HCEA** is responsible for receiving health certificates from a **health registration authority (HRA)**. This term can also be used to refer to the client machine in the Health Certificate Enrollment Protocol.

health registration authority (HRA): The server-side component in the Health Certificate Enrollment Protocol. The **HRA** is a registration authority (RA) that requests a health certificate from a **certification authority (CA)** upon validation of health.

language code identifier (LCID): A 32-bit number that identifies the user interface human language dialect or variation that is supported by an application or a client computer.

Lightweight Directory Access Protocol (LDAP): The primary access protocol for Active Directory. Lightweight Directory Access Protocol (LDAP) is an industry-standard protocol, established by the Internet Engineering Task Force (IETF), which allows users to query and update information in a directory service (DS), as described in [MS-ADTS]. The Lightweight Directory Access Protocol can be either version 2 [\[RFC1777\]](#) or version 3 [\[RFC3377\]](#).

Network Access Protection (NAP): A feature of an operating system that provides a platform for system health-validated access to private networks. **NAP** provides a way of detecting the health state of a network client that is attempting to connect to or communicate on a network, and limiting the access of the network client until the health policy requirements have been met. **NAP** is implemented through quarantines and health checks, as specified in [\[TNC-IF-TNCCSPBSoh\]](#).

object identifier (OID): In the context of a directory service, a number identifying an object class or attribute. Object identifiers are issued by the ITU and form a hierarchy. An OID is represented as a dotted decimal string (for example, "1.2.3.4"). For more information on OIDs, see [\[X660\]](#) and [RFC3280] Appendix A. OIDs are used to uniquely identify certificate templates available to the **certification authority (CA)**. Within a certificate, OIDs are used to identify standard extensions, as described in [RFC3280] section 4.2.1.x, as well as non-standard extensions.

public key: One of a pair of keys used in public-key cryptography. The public key is distributed freely and published as part of a digital certificate. For an introduction to this concept, see [\[CRYPTO\]](#) section 1.8 and [\[IEEE1363\]](#) section 3.1.

Public Key Cryptography Standards (PKCS): A group of Public Key Cryptography Standards published by RSA Laboratories.

registry: A local system-defined database in which applications and system components store and retrieve configuration data. It is a hierarchical data store with lightly typed elements that are logically stored in tree format. Applications use the registry API to retrieve, modify, or delete registry data. The data stored in the registry varies according to the version of the operating system.

statement of health (SoH): A collection of data generated by a system health entity, as specified in [TNC-IF-TNCCSPBSoH], which defines the health state of a machine. The data is interpreted by a Health Policy Server, which determines whether the machine is healthy or unhealthy according to the policies defined by an administrator.

statement of health response (SoHR): A collection of data that represents the evaluation of the **statement of health (SoH)** according to network policies, as specified in [TNC-IF-TNCCSPBSoH].

system health agent (SHA): The client components that make declarations on a specific aspect of the client health state and generate a statement of health ReportEntry (SoH ReportEntry).

tool extension GUID or administrative plug-in GUID: A GUID defined separately for each of the user policy settings and computer policy settings that associates a specific administrative tool plug-in with a set of policy settings that can be stored in a **Group Policy Object (GPO)**.

Unicode: A character encoding standard developed by the Unicode Consortium that represents almost all of the written languages of the world. The **Unicode** standard [\[UNICODE5.0.0/2007\]](#) provides three forms (UTF-8, UTF-16, and UTF-32) and seven schemes (UTF-8, UTF-16, UTF-16 BE, UTF-16 LE, UTF-32, UTF-32 LE, and UTF-32 BE).

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[MS-DHCPN] Microsoft Corporation, "[Dynamic Host Configuration Protocol \(DHCP\) Extensions for Network Access Protection \(NAP\)](#)".

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-GPOL] Microsoft Corporation, "[Group Policy: Core Protocol](#)".

[MS-GPREG] Microsoft Corporation, "[Group Policy: Registry Extension Encoding](#)".

[MS-HCEP] Microsoft Corporation, "[Health Certificate Enrollment Protocol](#)".

[MS-LCID] Microsoft Corporation, "[Windows Language Code Identifier \(LCID\) Reference](#)".

[MS-PEAP] Microsoft Corporation, "[Protected Extensible Authentication Protocol \(PEAP\)](#)".

[MS-TSGU] Microsoft Corporation, "[Terminal Services Gateway Server Protocol](#)".

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC2616] Fielding, R., Gettys, J., Mogul, J., et al., "Hypertext Transfer Protocol -- HTTP/1.1", RFC 2616, June 1999, <https://www.rfc-editor.org/info/rfc2616>

[RFC2782] Gulbrandsen, A., Vixie, P., and Esibov, L., "A DNS RR for specifying the location of services (DNS SRV)", RFC 2782, February 2000, <https://www.rfc-editor.org/info/rfc2782>

[RFC2818] Rescorla, E., "HTTP Over TLS", RFC 2818, May 2000, <https://www.rfc-editor.org/info/rfc2818>

[RFC2986] Nystrom, M. and Kaliski, B., "PKCS#10: Certificate Request Syntax Specification", RFC 2986, November 2000, <http://www.rfc-editor.org/info/rfc2986>

[RFC3174] Eastlake III, D., and Jones, P., "US Secure Hash Algorithm 1 (SHA1)", RFC 3174, September 2001, <https://www.rfc-editor.org/info/rfc3174>

[RFC3447] Jonsson, J. and Kaliski, B., "Public-Key Cryptography Standards (PKCS) #1: RSA Cryptography Specifications Version 2.1", RFC 3447, February 2003, <https://www.rfc-editor.org/info/rfc3447>

[TNC-IF-TNCCSPBSoh] TCG, "TNC IF-TNCCS: Protocol Bindings for SoH", version 1.0, May 2007, <https://trustedcomputinggroup.org/tnc-if-tnccs-protocol-bindings-soh/>

1.2.2 Informative References

[MS-NAPOD] Microsoft Corporation, "[Network Access Protection Protocols Overview](#)".

[MSDN-ALG] Microsoft Corporation, "CNG Algorithm Identifiers", [http://msdn.microsoft.com/en-us/library/aa375534\(VS.85\).aspx](http://msdn.microsoft.com/en-us/library/aa375534(VS.85).aspx)

[MSDN-CSP] Microsoft Corporation, "Cryptographic Provider Names", <http://msdn.microsoft.com/en-us/library/aa380243.aspx>

[MSDN-DHCP] Microsoft Corporation, "Dynamic Host Configuration Protocol", <http://technet.microsoft.com/en-us/network/bb643151.aspx>

[MSDN-NAP] Microsoft Corporation, "Network Access Protection", [http://msdn.microsoft.com/en-us/library/aa369712\(VS.85\).aspx](http://msdn.microsoft.com/en-us/library/aa369712(VS.85).aspx)

[MSDN-RAS] Microsoft Corporation, "RASENTRY structure", <http://msdn.microsoft.com/en-us/library/aa377274.aspx>

[MSDN-SC] Microsoft Corporation, "Smart Card Minidrivers", <https://learn.microsoft.com/en-us/windows-hardware/drivers/smartcard/smart-card-minidrivers>

[MSFT-IPSEC] Microsoft Corporation, "IPsec", <http://technet.microsoft.com/en-us/network/bb531150.aspx>

[MSFT-NAPIPSEC] Microsoft Corporation, "IPsec Enforcement Configuration", [http://technet.microsoft.com/en-us/library/dd125312\(WS.10\).aspx](http://technet.microsoft.com/en-us/library/dd125312(WS.10).aspx)

[MSFT-RDG] Microsoft Corporation, "Configuring the TS Gateway NAP Scenario", [http://technet.microsoft.com/en-us/library/cc732172\(WS.10\).aspx](http://technet.microsoft.com/en-us/library/cc732172(WS.10).aspx)

1.3 Overview

Network Access Protection (NAP) is a platform that controls access to network resources, based on a client computer's identity and compliance with corporate governance policy. NAP allows network administrators to define granular levels of network access, based on who a client is, the groups to which the client belongs, and the degree to which that client is compliant with corporate governance policy. Based on the degree of compliance, NAP can implement different enforcement methods that can restrict or limit client access to the network. If a client is not compliant, NAP provides a mechanism to automatically bring the client back into compliance and then to dynamically increase its level of network access. The NAP architecture is specified in [\[MS-NAPOD\]](#).

The behavior of NAP can be controlled through **Group Policy** by updating the client **registry**, as specified in [\[MS-GPOL\]](#) and in [\[MS-GPREG\]](#). This mechanism can be used by an administrator to enable or disable NAP enforcement, to set **Health Registration Authorities (HRAs)** to be used by the client, and to control client user interface and tracing. All NAP group policies are machine-specific, meaning that the same policy is applied to all users on a given machine.

1.3.1 Background

The Group Policy: Core Protocol, as specified in [\[MS-GPOL\]](#), allows clients to discover and retrieve policy settings created by administrators of a **domain**. These settings are persisted within Group Policy Objects (GPOs) assigned to policy target accounts, which are either computer accounts or user accounts in Active Directory. Each client uses the **Lightweight Directory Access Protocol (LDAP)** to determine which GPOs are applicable to it by consulting the Active Directory objects corresponding to its computer account and the user accounts of any users that log on to the client computer.

On each client, each GPO is interpreted and acted upon by software components known as client-side plug-ins. Each client-side plug-in is associated with a specific class of settings. The client-side plug-ins that are responsible for a given GPO are specified by using an attribute on the GPO. This attribute specifies a list of **GUID** pairs. The first GUID of each pair is referred to as a **client-side extension GUID (CSE GUID)**. The second GUID of each pair is referred to as a **tool extension GUID**.

For each GPO that is applicable to a client, the client consults the CSE GUIDs listed in the GPO to determine which client-side plug-ins on the client will handle the GPO. The client then invokes the client-side plug-ins to handle the GPO. Next, the client-side plug-in uses the contents of the GPO to retrieve and process settings specific to its class, in a manner specific to the plug-in.

1.3.2 Group Policy Extension Overview

NAP client configuration Group Policy settings are accessible from a **GPO** through the Group Policy: NAP Extension to the Group Policy: Core Protocol. The extension provides a mechanism for administrative tools to obtain metadata about **registry**-based settings.

The process of configuring and applying the NAP Group Policy settings consists of the following steps:

1. An administrator invokes a Group Policy administrative tool to administer the NAP client configuration settings through the Group Policy: NAP Extension. The NAP Extension reads and updates a generic settings database using the Group Policy: Registry Extension Encoding, as specified in [\[MS-GPREG\]](#) section 3.1.5.8, which results in the storage and retrieval of settings on a **Group Policy server**. These settings describe configuration parameters to be applied to a generic settings database on a client that is affected by the GPO.

The administrator views the data and updates it as desired.

2. A client computer affected by that GPO is started (or is connected to the network, if this happens after the client starts), and the Group Policy: Core Protocol is invoked by the client to retrieve Policy Settings from the Group Policy server. As part of this processing, the registry extension's **CSE GUID** (as specified in [\[MS-GPREG\]](#) section 1.9) is read from the GPO.

3. The presence of the registry extension's CSE GUID (as specified in [MS-GPREG] section 1.9) in the GPO instructs the client to invoke a registry extension plug-in component for policy application. This component parses the file of settings and saves them in the generic settings database (registry) on the local machine.
4. The NAP subsystem on the client recognizes that its configuration has been updated and takes the appropriate actions.

This document specifies the behavior of the administrative plug-in mentioned in step 1. The operation of the Group Policy: Core Protocol in step 2 is specified in [MS-GPOL] section 3.2. The process of retrieving the settings in step 3 is specified in [MS-GPREG] section 3.2. Step 4 is specific to a NAP client implementation.

1.4 Relationship to Protocols and Other Structures

Configuration changes updated on the **Group Policy server** are dependent on the Group Policy: Registry Extension Encoding, as specified in [MS-GPREG] section 3.1.5.8 (and all protocols specified in [MS-GPREG] section 1.4), which reads the Group Policy: NAP Extension data structure and updates the registry.pol file on the Group Policy server.

The distribution of the Group Policy: NAP Extension data structure to the client is dependent on the Group Policy: Registry Extension Encoding, as specified in [MS-GPREG] (and all protocols specified in [MS-GPREG] section 1.4), which retrieves settings from a **GPO** and populates those settings in the client **registry**.

The Group Policy: Registry Extension Encoding as specified in [MS-GPREG] is invoked as an extension of the Group Policy: Core Protocol, as specified in [MS-GPOL].

The generic settings database<1> on the local machine maintains the local configuration, which is used by the NAP in case the local machine does not participate in Group Policy.<2>

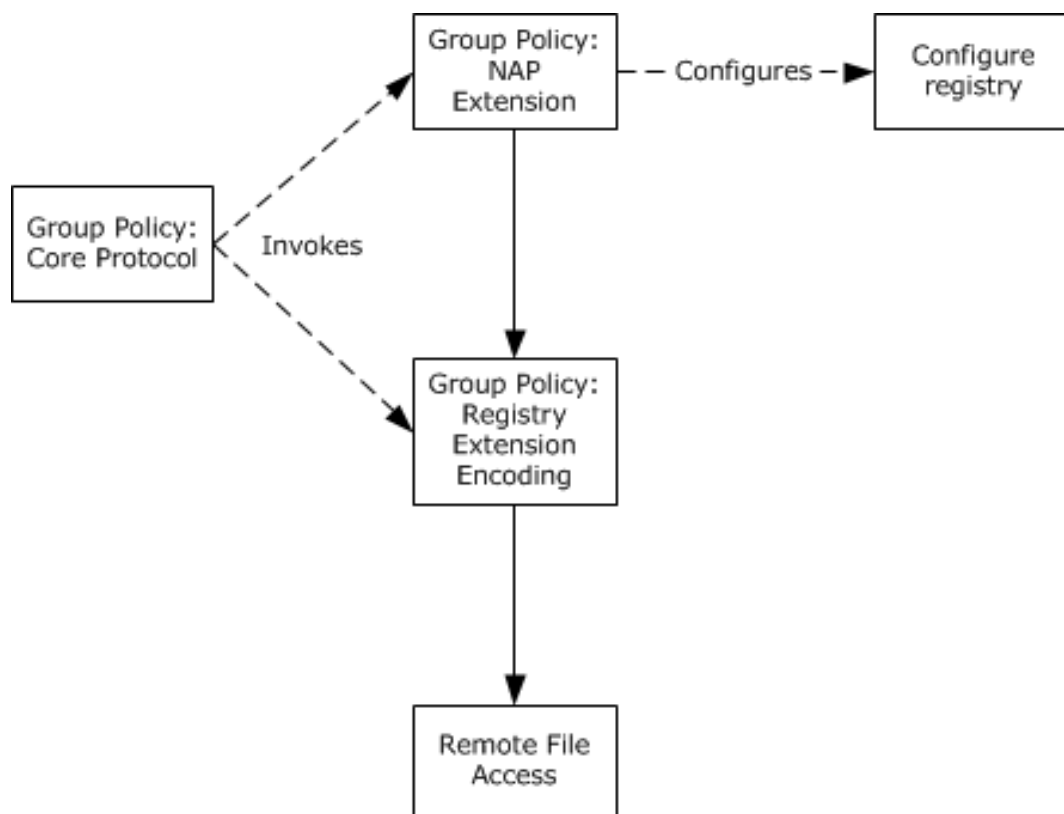


Figure 1: Protocol relationship diagram

1.5 Applicability Statement

The Group Policy: NAP Extension is applicable only within the Group Policy framework and is used to configure certain aspects of NAP behavior on such clients.

The Group Policy: NAP Extension has only an administrative-side extension and no client-side extension. This extension updates the generic settings database<3> and is documented here for informative purposes only.

1.6 Versioning and Localization

This document covers versioning issues in the following areas:

- Structure Versions: There is no versioning mechanism in the Group Policy: NAP Extension. If new functionality is required that would be incompatible with the existing **registry** settings, a new Group Policy extension is implemented.
- Localization: Localization-dependent registry content is specified in section 2.2.

The Group Policy: NAP client configuration contains registry keys with data that is used for display purposes; these keys can contain localization-dependent content.

1.7 Vendor-Extensible Fields

The Group Policy: NAP Extension does not define any vendor-extensible fields.

This structure defines **tool extension GUID** values, as specified in [MS-GPOL] section 1.8. The following assignment represents the **GUID** in string format.

Parameter	Value
Tool extension GUID (computer policy settings)	{A2A54893-AAF2-49A3-B3F5-CC43CEBCC27C}

2 Structures

This protocol references commonly used data types as defined in [\[MS-DTYP\]](#). The NAP Group Policy administrative plug-in uses the transport specified in [\[MS-GPOL\]](#) to read and modify settings in the central policy store. Information is retrieved from the policy store by the Group Policy: Registry Extension Encoding ([\[MS-GPREG\]](#) section 3.2), which writes the information to the client's **registry**.

The NAP Group Policy client configuration is implemented as a set of entries in the machine-specific Registry Policy file used by the Group Policy: Registry Extension Encoding. To support the NAP Group Policy option, the NAP administrative plug-in MUST be able to write and query the corresponding entry in the machine-specific Registry Policy file of the relevant GPO.

The following NAP Group Policy client configurations are defined:

- Trace settings
- User interface settings
- Enforcement client settings
- **Health registration authority (HRA)** settings
- **Statement of health (SoH)** settings

The following sections specify the format of the corresponding entries in the machine-specific Registry Policy file. The intent of various settings is also described in the following sections; however, these settings are processed by the NAP system in the client, and their descriptions here are only for informative purposes, not for normative purposes.

2.1 Trace Settings

The NAP client tracing functionality settings are compounded from two **registry** entries that are represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\

2.1.1 Enable Tracing

Value: "Enable Tracing" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to the size of the Data field.

Data: A 32-bit unsigned integer.

Value	Meaning
0x00000000	Disables NAP tracing on the client.
0x00000001	Enables NAP tracing on the client.

2.1.2 Tracing Level

Value: "Tracing Level" or one of the value names listed in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit unsigned integer.

Value	Meaning
0x00000000	Disables NAP tracing on the client.
0x00000001	Sets NAP tracing on the client to basic level.
0x00000002	Sets NAP tracing on the client to advanced level.
0x00000003	Sets NAP tracing on the client to debug level.

2.2 User Interface Settings

The NAP client uses user interface **registry** content as display information when user interface registry keys are available; otherwise, the user interface will not display a title and description. The user interface registry keys can contain localization-dependent content. User interface registry entries can be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\UI\<LCID>

The <LCID> **language code identifier (LCID)** values are specified in [\[MS-LCID\]](#) section 2.2.

2.2.1 SmallText

Value: "SmallText" or one of the value names listed in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_SZ.

Size: Equal to size of the Data field.

Data: A variable-length null-terminated **Unicode** string. This setting specifies the user notification title displayed to the user.

2.2.2 LargeText

Value: "LargeText" or one of the value names listed in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_SZ.

Size: Equal to the size of the Data field.

Data: A variable-length null-terminated **Unicode** string. This setting specifies the user notification sub-title displayed to the user.

2.2.3 ImageFile

The ImageFile entry can be represented in the machine-specific Registry Policy file as follows:

Value: "ImageFile" or one of the value names listed in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_BINARY.

Size: Equal to the size of the Data field.

Data: An octet stream representing an image that is displayed in the NAP client user interface. The data interpretation is determined by the image file name extension specified in [ImageFileName \(section 2.2.4\)](#).

NAP client implementations interpret this setting as the company logo to display. If this key is missing, no image is displayed. Implementations MAY choose to support this option. If this option is supported, the image file name key (see ImageFileName (section 2.2.4) MUST be available.

2.2.4 ImageFileName

Value: "ImageFileName" or one of the value names listed in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_SZ.

Size: Equal to size of the Data field.

Data: A variable-length null-terminated **Unicode** string. This setting is used to determine the format of the image data specified in section [2.2.3](#). File formats, as indicated in the file type extension, include but are not restricted to bmp, icon, gif, jpeg, exif, png, tiff, wmf, or emf.

2.3 Enforcement Client Settings

A NAP **enforcement client** uses the health state of a computer to request a certain level of access to a network. This is done using NAP protocol **SoH** ([\[TNC-IF-TNCCSPBSOH\]](#) section 3.5) and **statement of health response (SoHR)** ([\[TNC-IF-TNCCSPBSOH\]](#) section 3.6) messages exchanged between a client and a server to validate client conformance with corporate security policies.

Different types of mechanisms transport SoHs intended to manage the health of connected resources. These mechanisms, called enforcement clients, are configured from the NAP Group Policy and are listed in the following table.

Enforcement client	<qec-id> value	Description
Dynamic Host Configuration Protocol (DHCP)	79617	Enforces health policies when a client computer attempts to obtain an IP address from a DHCP server. The implementation is specified in section 2.3.1 .
Remote access	79618	Enforces health policies when a client computer attempts to gain access to the network through a virtual private network (VPN) connection. The implementation is specified in section 2.3.2 .
Internet Protocol security (IPsec)	79619	Enforces health policies when a client computer attempts to communicate with another computer using IPsec. The implementation is specified in section 2.3.3 .
Wireless EAPOL	79620	Enforces health policies when a client computer attempts to access a

Enforcement client	<qec-id> value	Description
		network through an 802.1X wireless connection or an authenticating switch connection. 4 The implementation is specified in section 2.3.5 .
Remote desktop gateway (RDG)	79621	Enforces health policies when a client computer attempts to gain access to an RDG. The implementation is specified in section 2.3.4 .
Extensible Authentication Protocol (EAP)	79623	Enforces health policies when a client computer attempts to access a network through an 802.1X wireless connection or an authenticating switch connection. The implementation is specified in section 2.3.5 .

For more information on NAP enforcement clients, see [\[MSDN-NAP\]](#).

The NAP enforcement client settings are compounded from one **registry** entry per enforcement client that **MUST** be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Qecs\<qec-id>

All the <qec-id> keys **MUST** have the following value:

Value: "Enabled" or one of the value names listed in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit unsigned integer.

Value	Meaning
0x00000000	Disables NAP enforcement.
0x00000001	Enables NAP enforcement.

2.3.1 DHCP Enforcement

The **Dynamic Host Configuration Protocol (DHCP)** NAP **enforcement client** provides functionality in the DHCP client service that uses industry standard DHCP messages to exchange system health messages specified in section [2.3](#).

The client sends system health information to the DHCP enforcement server, using DHCP Extensions, as specified in [\[MS-DHCPN\]](#). The DHCP server can send the SoH to a policy server (for example NPS) for evaluation. Based on the policy server response, the DHCP enforcement server can provide IP addressing information that allows the client to connect to other computers, or it can provide IP addressing information that limits the computers to which the client can connect. Alternatively, the DHCP enforcement server might not provide IP addressing information.

For more information on DHCP, see [\[MSDN-DHCP\]](#).

2.3.2 Remote Access Enforcement

The remote access NAP **enforcement client** provides functionality in the **Remote Access Service (RAS)** that makes it possible to connect a remote client computer to a network server over a **virtual private network (VPN)** and to send health information provided by NAP.

When a client attempts to access a network over VPN, the VPN server can request an SoHR response message ([[TNC-IF-TNCCSPBSOH](#)] section 3.6) from the client by sending some PEAP TLV ([[MS-PEAP](#)]) messages. If the RAS enforcement client is enabled on the client, it responds with an SoH message, as specified in [MS-PEAP] section 2.2.8. The RAS server might send the SoH message to a policy server (for example NPS) for evaluation. Based on the policy server response, the RAS server can create a VPN connection that enables the client to connect to other computers on the network, or the RAS server can quarantine the client by limiting the computers to which it can connect using the VPN connection. Alternatively, the RAS server can reject the client access request.

For more information on RAS clients, see [[MSDN-RAS](#)].

2.3.3 IPsec Enforcement

The IPsec NAP **enforcement client** is a component that obtains an SoH, as specified in section 2.3, and sends it to a **HRA**. The protocol used by the client to communicate with the HRA is called the Health Certificate Enrollment Protocol (HCEP), as specified in [[MS-HCEP](#)].

The HCEP request message payload sent by the Health Certificate Enrollment Agent (HCEA) contains a **PKCS #10** certificate request, as specified in [[RFC2986](#)], which contains an SoH message ([[TNC-IF-TNCCSPBSOH](#)] section 3.5).

The HRA sends the SoH to a policy server (for example NPS) for evaluation. Based on the policy server response, the HRA performs the following steps:

- If the policy server response is that the client is compliant with corporate health policy, the HRA requests a **certificate authority (CA)** to issue a certificate and sends the issued certificate back to the client.
- If the client's health state is not compliant, the HRA can request a certificate from the certificate authority (CA), with the certificate containing an indication that the client is unhealthy.

In the IPsec scenario, the client is able to connect to a network, but the client does not have a valid health certificate for communicating on that network until the HCEP message exchange is completed. If the client is non-compliant, the HRA might not provide a health certificate.

The IPsec NAP enforcement client also interacts with the IPsec components to ensure that the health certificate is used for IPsec-protected communication.

For more information, see [[MSFT-IPSEC](#)] and [[MSFT-NAIPSEC](#)].

The IPsec NAP enforcement client can be configured to use the local predefined IPsec policy rather than the default IPsec policy. The use local IPsec policy **registry** entry can be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig

Value: "PlumbIpsecPolicy" or one of the value names specified in the table in [[MS-GPREG](#)] section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit unsigned integer.

Value	Meaning
0x00000000	Allows the use of domain-based IPsec Group Policy on the client.
0x00000001	Uses the local IPsec policy on the client.

2.3.4 RDG Enforcement

The NAP remote desktop gateway (RDG) **enforcement client** is a component that obtains the **SoH** from NAP as specified in section [2.3](#) and uses it while the client connects to a remote desktop server.

While attempting to access an RDG server, the remote desktop client (RDC) obtains an SoH, as specified in section 2.3, and sends it in the TSGU, as specified in [\[MS-TSGU\]](#).

The RDG server MAY send the SoH to a policy server (for example MPS) for evaluation. Based on the policy server response, the RDG server MAY grant access to the RDC, or the remote desktop server MAY grant access but deny access to certain machine resources such as hard drives, disks, PnP devices, and clipboards.[<5>](#)

For more information on RDG and NAP, see [\[MSFT-RDG\]](#).

2.3.5 EAP Enforcement

The NAP EAP **enforcement client** extends the 802.1x supplicant, allows responding to an **SoH** Request TLV message with an SoH TLV message, as specified in section [2.3](#), and sends the response using an 802.1x supplicant for 802.1x-authenticated connections, as described in [\[MS-NAPOD\]](#).[<6>](#)

While attempting to access a LAN or WLAN using an 802.1x connection, the 802.1x supplicant obtains an SoH as specified in section 2.3 and sends it in **PEAP-Type-Length-Value (TLV)** extension, as specified in [\[MS-PEAP\]](#) section 2.2.8. The 802.1x server can send the SoH to a policy server (for example NPS) for evaluation. Based on the policy server response, the 802.1x server can enable the client to connect to other computers on the network or can restrict the traffic of the NAP client by specifying a restricted network that limits access to specific resources on the network, as described in [\[MS-NAPOD\]](#). Alternatively, the 802.1x server can reject supplicant access.

2.4 Health Registration Authority (HRA) Settings

NAP **HRAs** are divided into groups that use the same certificate security settings specified in section [2.3.3](#). The HRA groups are represented in the **registry** as depicted in the following figure.

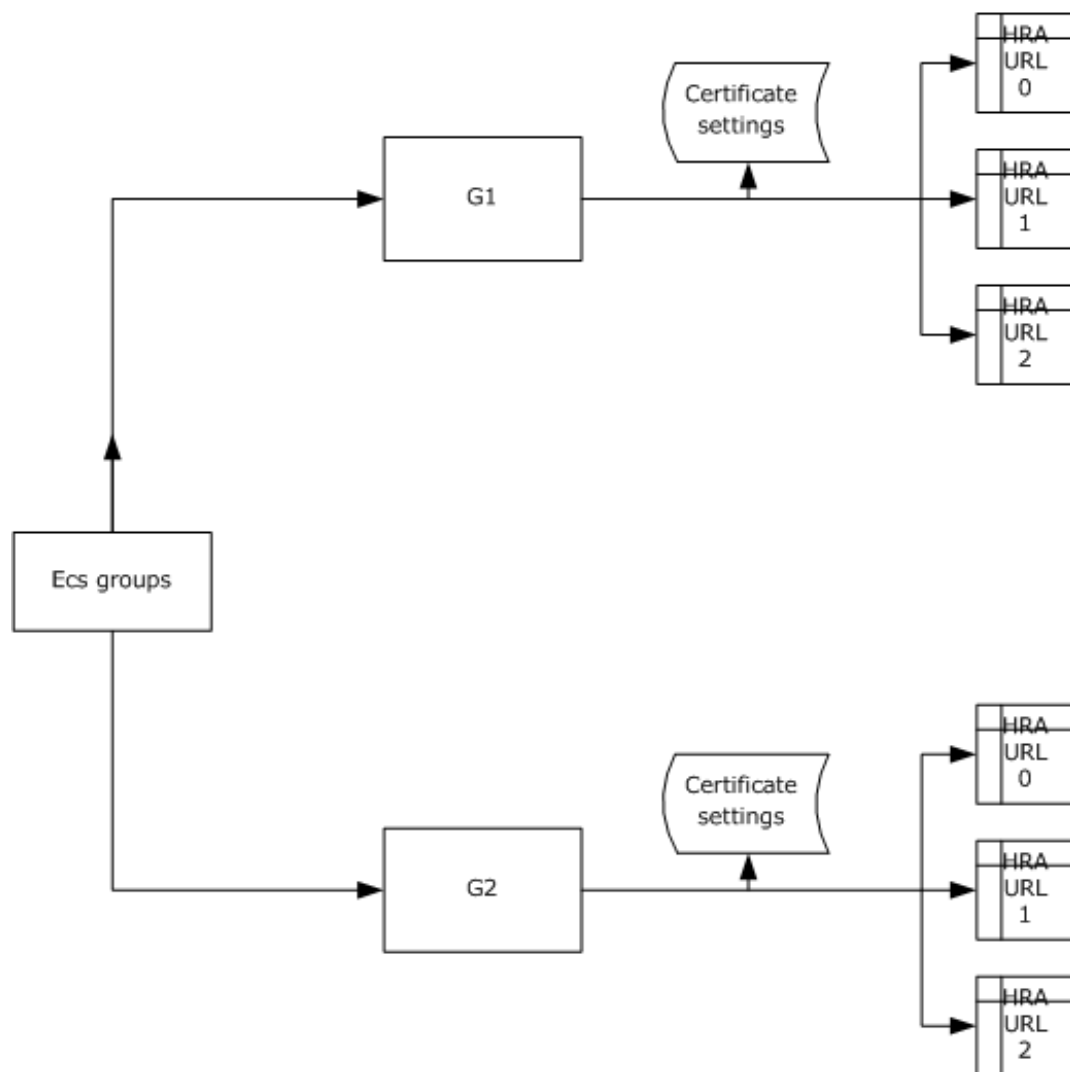


Figure 2: Health certificate server (hcs) groups registry representation

The NAP HRA settings are compounded from multiple registry entries that MUST be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Enroll\HcsGroups\<Server-Group>

<Server-Group> is the name of the HRAs group.

2.4.1 PKCS#10 Certificate Settings

The **health certificate enrollment agent (HCEA)** MUST be configured with the required security parameters to construct the **Public Key Cryptography Standards (PKCS)** #10 certificate request, as specified in [\[MS-HCEP\]](#) section 2.2.1.4.

These security parameters are configured in the NAP Group Policy. If the IPsec **enforcement client** is enabled, as specified in section [2.3](#), the security parameters entries MUST be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\ Enroll\HcsGroups\<Server-Group>

Security parameter values follow in sections [2.4.1.1](#) through [2.4.1.6](#).

2.4.1.1 Cryptographic Service Provider (CSP)

The name of the **cryptographic service provider (CSP)** used to generate the key pair on the HCEA.

Value: "CSP" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_SZ.

Size: Equal to size of the Data field.

Data: A variable-length null-terminated **Unicode** string. This setting specifies the name of the CSP used.

The following CSPs are available by default. [<7>](#)

CSP	Description
Microsoft Base Cryptographic Provider v1.0	A broad set of basic cryptographic functionality that can be exported to other countries or regions.
Microsoft Strong Cryptographic Provider	An extension of the Microsoft Base Cryptographic Provider.
Microsoft Enhanced Cryptographic Provider v1.0	Microsoft Base Cryptographic Provider with support for longer keys and additional algorithms.
Microsoft AES Cryptographic Provider	Microsoft Enhanced Cryptographic Provider with support for AES encryption algorithms.
Microsoft Base DSS Cryptographic Provider	Provides hashing, data signing, and signature verification capability, using the Secure Hash Algorithm 1 (SHA1) and Digital Signature Standard (DSS) algorithms.
Microsoft Base DSS and Diffie-Hellman Cryptographic Provider	A superset of the DSS Cryptographic Provider that also supports Diffie-Hellman key exchange, hashing, data signing, and signature verification, using the Secure Hash Algorithm 1 (SHA1) and Digital Signature Standard (DSS) algorithms.
Microsoft Enhanced DSS and Diffie-Hellman Cryptographic Provider	Supports Diffie-Hellman key exchange (a 40-bit DES derivative), SHA hashing, DSS data signing, and DSS signature verification.
Microsoft DH SChannel Cryptographic Provider	Supports hashing, data signing with DSS, generating Diffie-Hellman (D-H) keys, exchanging D-H keys, and exporting a D-H key. This CSP supports key derivation for the SSL3 and TLS1 protocols.
Microsoft RSA/SChannel Cryptographic Provider	Supports hashing, data signing, and signature verification. The algorithm identifier CALG_SSL3_SHAMD5 is used for SSL 3.0 and TLS 1.0 client authentication. This CSP supports key derivation for the SSL2, PCT1, SSL3, and TLS1 protocols.
Microsoft Base Smart Card Crypto Provider	Provides all of the functionality of the Microsoft Strong Cryptographic Provider . The Microsoft Base Smart Card Cryptographic Service Provider communicates with individual smart cards that translate the characteristics of particular smart cards into a uniform interface. For more information on smart cards, see [MSDN-SC] .
Microsoft Exchange Cryptographic Provider	A 64-bit block encryption CSP tied to the Mail API.

CSP	Description
v1.0	

2.4.1.2 Cryptographic Provider Type

The type of the **CSP** used to generate the key pair on the HCEA. There are many different standard data formats and protocols that CSP can use. These are generally organized into types, each of which has its own set of data formats and processing rules. For more information about CSP types, see [\[MSDN-CSP\]](#).

Value: "CSPTYPE" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit value consisting of the following type values.

Name	Value	Meaning
PROV_RSA_FULL	0x00000001	Supports both digital signatures and data encryption. It is considered a general purpose CSP. The RSA public key algorithm is used for all public key operations.
PROV_DSS	0x00000003	Supports hashes and digital signatures. The signature algorithm specified by the PROV_DSS provider type is the Digital Signature Algorithm (DSA) .
PROV_RSA_AES	0x00000018	Supports the same as PROV_RSA_FULL with additional AES encryption capability.
PROV_DSS_DH	0x0000000D	A superset of the PROV_DSS provider type with Diffie-Hellman key exchange.
PROV_DH_SCHANNEL	0x00000012	Supports both Diffie-Hellman and Schannel protocols.
PROV_RSA_SCHANNEL	0x0000000C	Supports both RSA and Schannel protocols.
PROV_MS_EXCHANGE	0x00000005	Designed for the cryptographic needs of the Exchange mail application and other applications compatible with Microsoft Mail.

2.4.1.3 Public Key OID

A **public key** OID is an **object identifier (OID)** identifying the algorithm of the public-private key pair associated with the certificate. For more information, see [\[RFC3447\]](#).

Value: "PublicKeyAlgOid" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_SZ.

Size: Equal to size of the Data field.

Data: A variable-length null-terminated **Unicode** string. This setting specifies the public key OID used.

The following table maps public key algorithm names and OIDs. For more information on the key algorithms, see [\[MSDN-ALG\]](#).

Name	OID
RSA	1.2.840.113549.1.1.1
DSA	1.2.840.10040.4.1
DH	1.2.840.10046.2.1
RSASSA-PSS	1.2.840.113549.1.1.10
DSA	1.3.14.3.2.12
DH	1.2.840.113549.1.3.1
RSA_KEYX	1.3.14.3.2.22
mosaicKMandUpdSig	2.16.840.1.101.2.1.1.20
ESDH	1.2.840.113549.1.9.16.3.5
NO_SIGN	1.3.6.1.5.5.7.6.2
ECC	1.2.840.10045.2.1
ECDSA_P256	1.2.840.10045.3.1.7
ECDSA_P384	1.3.132.0.34
ECDSA_P521	1.3.132.0.35
RSAES_OAEP	1.2.840.113549.1.1.7
ECDH_STD_SHA1_KDF	1.3.133.16.840.63.0.2

2.4.1.4 Public Key Length

The key length of the public-private key pair associated with the certificate. For more information, see [\[RFC3447\]](#).

Value: "KeyLength" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit value consisting of the public key length. The minimum "Public Key Length" expected is 0x00000800. If the "Public Key Length" is less than 0x00000800, the Data received from group policy is ignored, and the "Public Key Length" field is set to 0x00000800.

2.4.1.5 Public Key Spec

When a public-private key pair is generated, several types of keys can be created. Keys can be created to allow their use with encryption, digital signatures, or both. The **Value** represents the public key associated with the certificate.

Value: "PublicKeySpec" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted. [<8>](#)

Type: REG_DWORD.

Size: Equal to the size of the Data field.

Data: A 32-bit value that is set to 0x00000001 (AT_KEYEXCHANGE).

2.4.1.6 Hash Algorithm OID

The Hash Algorithm OID is an OID identifying the hash algorithm used to sign the certificate request. For more information on hash algorithms, see [\[RFC3174\]](#).

Value: "HashAlgOid" or one of the value names listed in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_SZ.

Size: Equal to size of the Data field.

Data: A variable-length null-terminated **Unicode** string. This setting specifies the public key OID used.

The list of supported hash algorithm OIDs follows.

Name	OID
sha1RSA	1.2.840.113549.1.1.5
md5RSA	1.2.840.113549.1.1.4
sha1DSA	1.2.840.10040.4.3
sha1RSA	1.3.14.3.2.29
shaRSA	1.3.14.3.2.15
md5RSA	1.3.14.3.2.3
md2RSA	1.2.840.113549.1.1.2
md4RSA	1.2.840.113549.1.1.3
md4RSA	1.3.14.3.2.2
md4RSA	1.3.14.3.2.4
md2RSA	1.3.14.7.2.3.1
sha1DSA	1.3.14.3.2.13
dsaSHA1	1.3.14.3.2.27
mosaicUpdatedSig	2.16.840.1.101.2.1.1.19

Name	OID
sha1NoSign	1.3.14.3.2.26
md5NoSign	1.2.840.113549.2.5
sha256NoSign	2.16.840.1.101.3.4.2.1
sha384NoSign	2.16.840.1.101.3.4.2.2
sha512NoSign	2.16.840.1.101.3.4.2.3
sha256RSA	1.2.840.113549.1.1.11
sha384RSA	1.2.840.113549.1.1.12
sha512RSA	1.2.840.113549.1.1.13
RSASSA-PSS	1.2.840.113549.1.1.10
sha1ECDSA	1.2.840.10045.4.1
sha256ECDSA	1.2.840.10045.4.3.2
sha384ECDSA	1.2.840.10045.4.3.3
sha512ECDSA	1.2.840.10045.4.3.4
specifiedECDSA	1.2.840.10045.4.3

2.4.2 HRA Auto-Discovery

HRA groups can be set by group policy or can be discovered automatically by the NAP client using DNS SRV lookup, as specified in [\[RFC2782\]](#). A NAP client discovers a suitable HRA at start-up using the following sequence:

1. Query SRV records for HRAs in the Active Directory site of the client (for example, _hra._tcp.<sitename>._sites.<domainname>)
2. Query SRV records for HRAs in the **Active Directory domain** of the client (for example, _hra._tcp.<domainname>)
3. Query SRV records for HRAs in the DNS **domain** of the client (for example, _hra._tcp.<DNSname>)

To enable HRA auto discovery, a **registry** setting entry MUST be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Enroll\HcsGroups

Value: "EnableDiscovery" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to the size of the Data field.

Data: A 32-bit unsigned integer.

Value	Meaning
0x00000000	Disables HRA auto discovery.
0x00000001	Enables HRA auto discovery.

2.4.3 Use SSL

The HCEP uses HTTP (as specified in [RFC2616](#)) or HTTP over TLS (as specified in [RFC2818](#)) as the transport for its messages. To configure how HCEP connects to the **HRA**, a registry setting entry MUST be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Enroll\HcsGroups\<Server-Group>

Value: "AllowNonSSL" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to the size of the Data field.

Data: A 32-bit unsigned integer.

Value	Meaning
0x00000000	Disables SSL.
0x00000001	Enables SSL.

Communication with the HRA is always performed using SSL when HRA auto-discovery is used; see section [2.4.1](#).

2.4.4 HRA URLs

Group Policy enables the administrator to configure specific **HRA** groups by setting the URL. To configure an HRA URL, a **registry** entry MUST be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Enroll\HcsGroups\<Server-Group>\<String>

<String> is the key name of a specific HRA. The Group Policy: NAP Extension uses Url#.

Url# is the string "Url" with the HRA order in the group concatenated to it. The Policy Group can define 0 to 254 HRAs in a group. If there are no URLs defined and HRA auto-discovery specified in section [2.4.2](#) is set to 0 the NAP client won't be able to request a health certificate as specified in section [2.3.3](#).

Example: A URL of the first HRA is defined under the registry key "Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Enroll\HcsGroups\MyGroup\Url0".

2.4.4.1 Server

Value: "Server" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_SZ.

Size: Equal to size of the Data field.

Data: A variable-length null-terminated **Unicode** string. This setting specifies the **HRA** URL to connect to.

2.4.4.2 Order

Value: "Order" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit value in the range from 0x00000000 to 0x000000FE, inclusive.

2.4.5 Reconnect Attempts

Group Policy enables the administrator to configure how long the client waits before attempting to reconnect to an **HRA** in the event of a connection failure. To configure a blackout interval, a **registry** entry **MUST** be represented in the machine-specific Registry Policy file as follows:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Enroll\HcsGroups\<Server-Group>

Value: "BlackOutIntervalInMinutes" or one of the value names listed in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit value representing time in minutes. For example, 0x0000000A represents 10 minutes of blackout.

2.5 SoH Settings

SoH specified in [\[TNC-IF-TNCCSPBSoH\]](#) has two settings that an administrator can configure using Group Policy. These settings are represented in the machine-specific registry as values ShatimeoutInMsec (section [2.5.1](#)) and BackwardCompatible (section [2.5.2](#)). In case these registry values are not set by the administrator, the settings are assigned default data by the NAP client. To configure SoH, **registry** values ShatimeoutInMsec and BackwardCompatible can be present in the machine-specific Registry Policy file under the following key:

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\

2.5.1 Task Timer

A Task Timer is a **system health agent (SHA)** timeout. A task timer is associated with all function calls that a **SoH** client makes to the SHA. The SHA is expected to complete the call within the timeout. Otherwise, the call is canceled and an error is reported by the SoH client.

Value: "ShatimeoutInMsec" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit value representing time in milliseconds. For example, 0x0000000A represents 10 milliseconds of blackout.

2.5.2 Backward Compatible

This setting determines the version of the message content, as specified in [\[TNC-IF-TNCCSPBSoh\]](#) section 3.5.1.1.

Value: "BackwardCompatible" or one of the value names specified in the table in [\[MS-GPREG\]](#) section 3.2.5.1 specifying how the value is deleted.

Type: REG_DWORD.

Size: Equal to size of the Data field.

Data: A 32-bit value consisting of the following values.

Value	Meaning
0x00000001	SSoH ([TNC-IF-TNCCSPBSoh] section 3.5.1.3) SoHReportEntry (0 plus)
0x00000002	SoH Mode Subheader SSoH ([TNC-IF-TNCCSPBSoh] section 3.5.1.3) SoHReportEntry (0 plus)

3 Structure Examples

In the following example, an administrator sets up a new **domain** and attempts to enable NAP DHCP enforcement on the computers in the domain. The client computers run operating systems that contain NAP client processes initialized at startup and terminated at shutdown.

First, the administrator installs and configures an operating system on a computer that is intended to function as the **domain controller (DC)**. After taking the necessary steps to designate the computer as a DC and creating a user account over the new domain, the administrator restarts the machine and logs on as the newly created user.

Next, the administrator launches the user interface for the administrative plug-in and sets the DHCP enforcement to Enabled. This causes the following entry to be written to the machine-specific Registry Policy file of the relevant GPO.

Key: Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Qecs\79617

Value: "Enabled".

Type: REG_DWORD.

Size: Equal to the size of the Data field.

Data: 0x00000001.

The administrator then adds client computers to this domain. When a client computer is restarted for the first time after being added to the domain, it contacts the domain controller (DC) and reads Group Policy information, as specified in [\[MS-GPOL\]](#). As part of this process, a machine-specific registry policy file containing the following items is also downloaded:

- A set of values under the **registry** key Software\Policies\Microsoft\NetworkAccessProtection\ClientConfig\Qecs\79617 that indicates that the NAP client will instruct the DHCP client on the system to send an SoH when requesting the IP address for the machine, as specified in section [2.3.1](#).

The Group Policy: Registry Extension Encoding on the client parses this file and adds the configuration information to the machine's registry.

The NAP client process polls the registry and determines that its Group Policy settings have changed. The NAP process then reads the enforcement values and sets the system DHCP client to send an SoH.

When a user logs on to the computer, the DHCP client requests the NAP agent for an SoH. NAP invokes the SHA to collect health information and to generate an SoH. The SoH is then sent by the DHCP client to the policy server.

The following figure represents such a transaction.

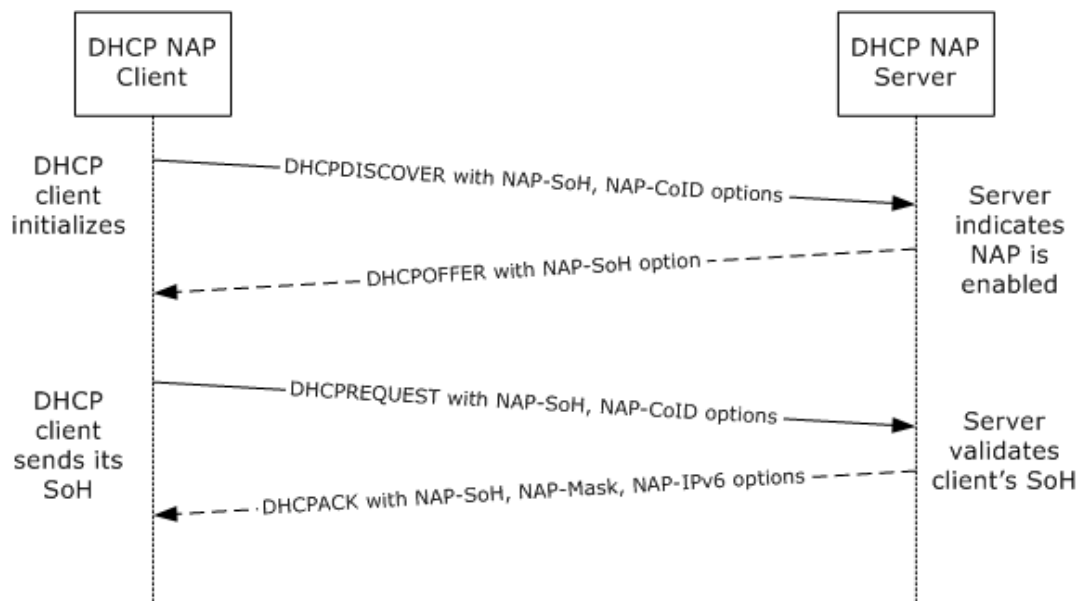


Figure 3: DHCP new lease acquisition process

When the SoH is sent, the client requests access to a service and, as a precondition for that access, is required to prove that it is in good health. When the SoH is received, it is forwarded to an infrastructure server that evaluates the SoH and returns the response (the SoHR) to the client by means of the original receiver of the SoH.

Generally, the receipt of an SoHR by the client allows access to the service being requested. When the health of the client is not good, the SoHR is likely to contain sufficient instructions to allow the client to seek and receive remedy. After the client is restored to good health, the client can initiate the protocol again.

4 Security

4.1 Security Considerations for Implementers

The Group Policy: NAP Extension sets the **NAP** enforcement policy on the client computer. This policy consists of **HRA** URLs and HRA connection transport and certificate security settings, as well as enforcement enabling. These configurations can also be set by the user through the NAP configuration UI. Therefore, it is extremely important that an implementation provide a means of protecting the integrity of the NAP policy against tampering, especially during its transfer from server to client. Ideally, this implementation-specific security method is provided as part of the transport for the Group Policy: Core Protocol.

4.2 Index of Security Fields

None.

5 Appendix A: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows 2000 operating system
- Windows XP operating system
- Windows Server 2003 operating system
- Windows Server 2003 R2 operating system
- Windows Vista operating system
- Windows Server 2008 operating system
- Windows 7 operating system
- Windows Server 2008 R2 operating system
- Windows 8 operating system
- Windows Server 2012 operating system
- Windows 8.1 operating system
- Windows Server 2012 R2 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 1.4](#): Windows implements the generic settings database using the **registry**.

[<2> Section 1.4](#): Windows maintains the local configuration under the [HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\NapAgent] registry key.

[<3> Section 1.5](#): Windows implements the generic settings database using the registry.

[<4> Section 2.3](#): The wireless EAPOL enforcement client is available only on NAP client computers running Windows XP operating system Service Pack 3 (SP3).

[<5> Section 2.3.4](#): The Remote Desktop Gateway Enforcement Client is available with the following Windows versions:

- Windows XP SP3 with Remote Desktop Connection (Terminal Services Client 6.0) installed
- Windows Vista
- Windows 7
- Windows 8
- Windows 8.1

[<6> Section 2.3.5](#): EAP enforcement is available in the following Windows versions:

- Windows Vista
- Windows 7
- Windows 8
- Windows 8.1

[<7> Section 2.4.1.1](#): For more information about the CSPs that are available on Windows, see [\[MSDN-CSP\]](#).

[<8> Section 2.4.1.5](#): The value "PublicKeySpec" is supported in Windows 7, Windows 8, and Windows 8.1. In Windows Vista and Windows XP SP3, the value is named "KeyType".

6 Change Tracking

No table of changes is available. The document is either new or has had no changes since its last release.

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